Event Management System — Test Cases (6 per Functional Requirement)

FR-001 — Account Registration & Email Verification

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	User Registration	Verify account is created with valid signup details.	User is on the Sign Up page.	1: Enter name, email, password. 2: Click Sign Up.	Name: Amit Rao; Email: amit.rao@test.com; Password: Pass@123	Account is created and shown as 'pending verification'; login is blocked until verified.	Created, verified=false (pending).	Pass
UT_02	User Registration	Verify signup is rejected for an invalid email format.	User is on the Sign Up page.	1: Enter name, password. 2: Enter malformed email. 3: Click Sign Up.	Name: Amit Rao; Email: amitrao@@test; Password: Pass@123	Signup is rejected with an 'invalid email format' error; no account is created.	400 invalid email error; no account created.	Pass
IT_01	Email Verification	Verify clicking the emailed verification link activates the account.	A pending account exists with a verification email sent.	1: Open verification email. 2: Click verification link. 3: Attempt login.	Verification link for amit.rao@test.com	Account status changes to 'active'; login succeeds immediately after verification.		
IT_02	Email Verification	Verify an expired/invalid verification link is rejected.	A verification link older than the configured expiry exists.	1: Click the expired verification link.	Expired verification link (>24h old)	System displays 'link expired/invalid' message; account remains inactive.		
ST_01	Registration Flow	Verify end-to-end signup, verification, and first login.	None.	1: Sign up with valid details. 2: Open email, click verification link. 3: Log in with the same credentials.	Name: Priya Nair; Email: priya.nair@test.com; Password: Nair@456	User completes signup, is verified, and logs in successfully with dashboard displayed.		
ST_02	Registration Flow	Verify signup is rejected when the email is already registered.	An active account already exists for the given email.	1: Enter name/password. 2: Enter an already-registered email. 3: Click Sign Up.	Email: amit.rao@test.com (existing)	Signup is rejected with 'email already registered' error; no duplicate account is created.	409 duplicate email error; no new row created.	Pass

FR-002 — Login & Password Reset

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Login	Verify login succeeds with correct credentials.	A verified, active account exists.	1: Enter email and password. 2: Click Login.	Email: amit.rao@test.com; Password: Pass@123	User is logged in and redirected to the dashboard within 5 seconds.		
UT_02	Login	Verify login is rejected with an incorrect password.	A verified, active account exists.	1: Enter correct email. 2: Enter wrong password. 3: Click Login.	Email: amit.rao@test.com; Password: WrongPass1	Login is rejected with an 'invalid credentials' error; no session is created.		
IT_01	Password Reset	Verify 'Forgot Password' sends a reset email with a valid link.	Account exists with a registered email.	1: Click 'Forgot Password'. 2: Enter registered email. 3: Submit.	Email: amit.rao@test.com	A time-limited password-reset email is sent to the registered address.		
IT_02	Password Reset	Verify the reset link updates the password and invalidates the old one.	A valid, unexpired reset link exists.	1: Open reset link. 2: Enter new password. 3: Submit. 4: Log in with old password.	New Password: NewPass@789	Password is updated; login with the old password fails; login with the new password succeeds.		
ST_01	Password Reset Flow	Verify complete forgot-password-to-login journey.	None.	1: Request reset. 2: Open email, click link. 3: Set new password. 4: Log in with new password.	Email: priya.nair@test.com; New Password: Nair@789	User successfully logs in using the newly set password.		
ST_02	Password Reset	Verify an expired reset link is rejected.	A reset link older than its expiry window exists.	1: Click the expired reset link. 2: Attempt to set a new password.	Expired reset link (>1h old)	System displays 'link expired' message; password is not changed.		

FR-003 — RBAC, Profile Updates & Admin Account Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Access Control	Verify a Participant cannot access the Organizer 'Create Event' page.	Logged in as a Participant.	1: Navigate directly to the Create Event URL/page.	Role: Participant	Access is denied with an authorization error; page content is not shown.		
UT_02	Profile Management	Verify a user can update their profile information.	User is logged in.	1: Go to Profile. 2: Edit contact number. 3: Save.	New Contact: 9876543210	Profile is updated and the new contact number is displayed on next view.		
IT_01	Admin Account Management	Verify admin suspending a user blocks that user's login.	Admin is logged in; target user account is active.	1: Admin opens user management. 2: Select target user. 3: Click Suspend.	Target user: amit.rao@test.com	User status changes to 'suspended'; subsequent login attempt by that user is blocked.		
IT_02	Admin Account Management	Verify admin deleting a user account removes login access.	Admin is logged in; target user account exists.	1: Admin opens user management. 2: Select target user. 3: Click Delete. 4: Confirm.	Target user: priya.nair@test.com	Account is removed; login attempt with deleted credentials fails.		
ST_01	Role-Based Navigation	Verify each role sees only its permitted menu items end-to-end.	Three accounts exist: one per role (Participant, Organizer, Admin).	1: Log in as Participant, note menu. 2: Log out, log in as Organizer, note menu. 3: Log out, log in as Admin, note menu.	Participant, Organizer, Admin credentials	Each role's menu shows only its permitted actions (e.g., only Admin sees 'Manage Users').		
ST_02	Admin Account Management	Verify a suspended user sees a clear error when attempting to log in.	Target account has been suspended by admin.	1: Attempt login with suspended account's credentials.	Suspended user: amit.rao@test.com	Login is rejected with an 'account suspended' message; no session/dashboard is shown.		

FR-004 — Event Creation & Update

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Event Creation	Verify an event is created when all required fields are valid.	Organizer is logged in.	1: Go to Create Event. 2: Fill name, description, date, time, venue, category, capacity. 3: Submit.	Name: Tech Fest; Date: 2026-10-10; Capacity: 100	Event is created and appears in the organizer's event list with all fields correct.		
UT_02	Event Creation	Verify event creation fails when a required field is missing.	Organizer is logged in.	1: Go to Create Event. 2: Leave 'capacity' blank. 3: Submit.	Name: Tech Fest; Capacity: (blank)	Submission is rejected with a 'required field missing' error; no event is created.		
IT_01	Event Update	Verify updating event details saves and reflects on the listing page.	An event created by the organizer exists.	1: Open the event. 2: Edit venue. 3: Save. 4: View event list.	New Venue: Auditorium B	Updated venue is saved and displayed immediately on the public event listing.		
IT_02	Event Creation	Verify the capacity field only accepts a positive integer.	Organizer is on the Create Event form.	1: Enter capacity as -5. 2: Submit.	Capacity: -5	Submission is rejected with an 'invalid capacity' error; no event is created.		
ST_01	Event Lifecycle	Verify an organizer can create then edit an event, with changes visible to participants end-to-end.	Organizer is logged in; Participant account exists separately.	1: Organizer creates event. 2: Organizer edits date. 3: Participant views event listing.	Name: Workshop A; New Date: 2026-11-01	Participant sees the updated date on the event listing without any stale data.		
ST_02	Access Control	Verify a Participant cannot create an event via direct page access.	Logged in as Participant.	1: Navigate to the Create Event page URL directly.	Role: Participant	Access is denied; Participant is redirected or shown an authorization error.		

FR-005 — Event Cancellation & Update Notifications

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Event Cancellation	Verify cancelling an event marks its status as cancelled.	An active event created by the organizer exists.	1: Open the event. 2: Click Cancel. 3: Confirm.	Event: Tech Fest	Event status changes to 'cancelled' and is flagged as such in the event list.		
UT_02	Event Cancellation	Verify a cancelled event rejects new registration attempts.	The target event has been cancelled.	1: Participant opens the cancelled event. 2: Click Register.	Event: Tech Fest (cancelled)	Registration is rejected with an 'event cancelled' message.		
IT_01	Notifications	Verify cancelling an event triggers a notification to all registrants.	At least 2 participants are registered for the event.	1: Organizer cancels the event. 2: Check notification queue/log for each registrant.	Registrants: 2 participants	A cancellation notification is dispatched to every registered participant within 5 minutes.		
IT_02	Notifications	Verify the cancellation notification content includes correct event details.	Event cancellation notification has been sent.	1: Open the received notification email.	Event: Tech Fest; Date: 2026-10-10	Email content correctly names the event and states it has been cancelled.		
ST_01	End-to-End Cancellation	Verify a registered participant receives the cancellation email within the defined window.	Participant is registered for an active event.	1: Organizer cancels the event. 2: Participant checks inbox within 5 minutes.	Event: Workshop A	Participant's inbox contains the cancellation email within 5 minutes of cancellation.		
ST_02	Event Cancellation	Verify cancelling an event with zero registrants completes without error.	Event exists with no registrations.	1: Organizer opens the event. 2: Click Cancel. 3: Confirm.	Event: Empty Seminar (0 registrants)	Event is cancelled successfully with no errors and no notifications sent.		

FR-006 — Event Discovery, Search, Filtering & Category Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Event Search	Verify searching by exact event name returns the matching event.	At least one event named 'Tech Fest' exists.	1: Enter 'Tech Fest' in the search bar. 2: Submit search.	Search term: Tech Fest	Search results contain the 'Tech Fest' event and no unrelated events.		
UT_02	Event Filtering	Verify filtering by category returns only matching events.	Events exist across multiple categories (e.g., Tech, Music).	1: Select category filter 'Music'. 2: Apply filter.	Category: Music	Only events tagged 'Music' are shown in the results.		
IT_01	Category Management	Verify an admin-added category appears in the organizer's event creation form.	Admin is logged in.	1: Admin adds category 'Workshops'. 2: Organizer opens Create Event form.	New category: Workshops	'Workshops' appears as a selectable category on the Create Event form.		
IT_02	Event Filtering	Verify filtering by a date range returns only events within that range.	Events exist with varying dates.	1: Set date filter from 2026-10-01 to 2026-10-15. 2: Apply filter.	Date range: 2026-10-01 to 2026-10-15	Only events falling within the specified date range are displayed.		
ST_01	Combined Search & Filter	Verify combined name search with category and date filters returns the correct intersection.	Multiple events exist with overlapping and non-overlapping attributes.	1: Enter search term. 2: Apply category filter. 3: Apply date filter. 4: Submit.	Search: Fest; Category: Tech; Date: Oct 2026	Only events matching all three criteria simultaneously are returned.		
ST_02	Event Listing	Verify cancelled events are excluded from listing and search results.	At least one cancelled event exists among active events.	1: View the event list. 2: Search for the cancelled event's name.	Cancelled event: Old Meetup	Cancelled event does not appear in the general listing or in search results.		

FR-007 — Event Registration & Duplicate Prevention

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Event Registration	Verify a valid registration creates a record with correct user, event, timestamp, and status.	Participant is logged in; event has available seats.	1: Open event. 2: Click Register. 3: Confirm.	Event: Tech Fest; User: amit.rao@test.com	A registration record is created with correct user, event, timestamp, and status 'confirmed'.		
UT_02	Duplicate Prevention	Verify a duplicate registration by the same user for the same event is rejected.	User is already registered for the event.	1: Open the same event again. 2: Click Register.	Event: Tech Fest; User: amit.rao@test.com (already registered)	Registration is rejected with an 'already registered' message; no duplicate record is created.		
IT_01	Seat Management	Verify a successful registration decrements the event's seat count in the database.	Event has N available seats before registration.	1: Note seat count. 2: Register for the event. 3: Recheck seat count.	Event: Workshop A; Seats before: 20	Seat count is exactly N-1 after the registration commits.		
IT_02	Organizer Reporting	Verify a new registration is retrievable in the organizer's registrant list.	Organizer owns the event that was just registered for.	1: Participant registers. 2: Organizer opens 'View Registrants'.	Event: Tech Fest	The newly registered participant appears in the organizer's registrant list.		
ST_01	End-to-End Registration	Verify a participant registers, sees confirmation, and appears in the organizer's list.	Participant and organizer accounts exist; event has open seats.	1: Participant registers for the event. 2: Participant sees confirmation screen. 3: Organizer views registrants.	Event: Tech Fest; User: priya.nair@test.com	Confirmation is shown to the participant and their entry appears in the organizer's registrant list.		
ST_02	Registration Validation	Verify registering for a cancelled or non-existent event is rejected.	Target event is cancelled or its ID no longer exists.	1: Attempt to register via the cancelled event's page/ID.	Event: Old Meetup (cancelled)	Registration is rejected with an appropriate error; no record is created.		

FR-008 — Seat Capacity Check & Concurrency Control

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Capacity Check	Verify registration is rejected when the event has zero available seats.	Event capacity is fully consumed (0 seats left).	1: Open the full event. 2: Click Register.	Event: Full Seminar (0 seats left)	Registration is rejected with an 'event full' message; seat count remains 0.		
UT_02	Capacity Check	Verify registration succeeds and decrements the seat count when a seat is available.	Event has exactly 1 seat remaining.	1: Register for the event.	Event: Almost Full Talk (1 seat left)	Registration succeeds; seat count becomes 0 immediately after.		
IT_01	Concurrency Control	Verify only one of two simultaneous requests for the last seat is confirmed.	Event has exactly 1 seat remaining.	1: Fire two registration requests for the same event at the same time (different users). 2: Check results.	Event: Almost Full Talk (1 seat left); Users: A, B	Exactly one request is confirmed; the other is rejected as 'event full'.		
IT_02	Concurrency Control	Verify seat capacity never goes negative under a simulated race condition.	Event has exactly 1 seat remaining.	1: Fire 10 simultaneous registration requests for the same event. 2: Check final seat count.	Event: Almost Full Talk (1 seat left); 10 concurrent requests	Final seat count is 0, never negative; exactly one registration is confirmed.		
ST_01	Load Test — Concurrency	Verify 100 concurrent requests for a 1-seat event yield exactly one confirmation.	Event has exactly 1 seat remaining; load-testing tool configured.	1: Fire 100 concurrent registration requests. 2: Count confirmations and rejections.	Event: Final Seat Test; 100 concurrent users	Exactly 1 confirmation and 99 rejections/waitlist offers are recorded; no overbooking occurs.		
ST_02	End-to-End Full Event	Verify a new participant sees 'Full' and is blocked from registering for a full event.	Event has 0 seats remaining.	1: Participant opens the full event page.	Event: Full Seminar (0 seats)	Page displays 'Event Full' and the Register button is disabled or leads to a waitlist offer.		

FR-009 — Waitlist, Self-Cancellation & Notifications

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Waitlist	Verify joining a waitlist records the user's position for a full event.	Event is full (0 seats); participant is not yet registered or waitlisted.	1: Open the full event. 2: Click 'Join Waitlist'.	Event: Full Seminar; User: amit.rao@test.com	User is added to the waitlist and shown their queue position (e.g., position 1).		
UT_02	Self-Cancellation	Verify a participant can cancel their own registration, freeing the seat.	Participant has an active registration for the event.	1: Open 'My Registrations'. 2: Click Cancel for the event. 3: Confirm.	Event: Tech Fest; User: priya.nair@test.com	Registration status changes to 'cancelled' and the event's seat count increases by 1.		
IT_01	Waitlist Auto-Promotion	Verify a cancellation automatically promotes the first waitlisted user.	Event is full; at least one user is on the waitlist; a registered participant cancels.	1: Registered participant cancels. 2: Check the waitlisted user's registration status.	Waitlist position 1: amit.rao@test.com	First waitlisted user's status changes to 'confirmed' and they receive a notification.		
IT_02	Notifications	Verify a confirmation email is sent immediately after a successful registration.	Participant successfully registers for an event with open seats.	1: Register for the event. 2: Check the participant's inbox.	Event: Workshop A; User: priya.nair@test.com	A registration confirmation email is received within 5 minutes.		
ST_01	End-to-End Waitlist Flow	Verify a waitlisted user is auto-promoted and notified when a seat frees up.	Event is full; Participant X is on the waitlist; Participant Y is registered.	1: Participant X joins waitlist (event full). 2: Participant Y cancels registration. 3: Check Participant X's status and inbox.	Waitlisted: Participant X; Cancelling: Participant Y	Participant X's status becomes 'confirmed' and they receive a promotion notification.		
ST_02	Reminder Notifications	Verify a reminder notification is sent to registered participants before the event date.	Participant is registered for an event starting within the reminder window.	1: Advance/trigger the scheduled reminder job for an event starting in 24 hours. 2: Check participant's inbox.	Event date: T+24h; Reminder window: 24h	Participant receives a reminder email referencing the correct event and time.		

FR-010 — Reporting, Export & Platform Statistics

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	Reporting	Verify the displayed registered-participant count matches actual active registrations.	Event has a known number of active registrations (e.g., 15).	1: Open the event page. 2: Check the displayed participant count.	Event: Tech Fest; Active registrations: 15	Displayed count reads exactly 15, matching the database.		
UT_02	Reporting	Verify the displayed available-seat count equals capacity minus active registrations.	Event capacity is 100 with 15 active registrations.	1: Open the event page. 2: Check the displayed seats-available figure.	Capacity: 100; Active registrations: 15	Displayed available seats reads exactly 85.		
IT_01	Participant Export	Verify the exported CSV participant list matches the on-screen registrant list.	Organizer owns an event with multiple registrants.	1: Open 'View Registrants'. 2: Click 'Export CSV'. 3: Open the downloaded file.	Event: Tech Fest; Registrants: 15	CSV contains all 15 registrants with fields matching the on-screen list exactly.		
IT_02	Participant Export	Verify the exported PDF participant list matches the on-screen registrant list.	Organizer owns an event with multiple registrants.	1: Open 'View Registrants'. 2: Click 'Export PDF'. 3: Open the downloaded file.	Event: Tech Fest; Registrants: 15	PDF contains all 15 registrants with fields matching the on-screen list exactly.		
ST_01	Platform Statistics	Verify the admin statistics page shows correct totals for events, users, and registrations.	Admin is logged in; database has known counts of events/users/registrations.	1: Admin opens the Platform Statistics page.	Known totals: 50 events, 300 users, 900 registrations	Displayed totals exactly match the known database counts.		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_02	Organizer Reporting	Verify an organizer can view accurate event-wise registration info across multiple events.	Organizer owns 3 events with differing registration counts.	1: Open 'My Events'. 2: View registration info for each of the 3 events.	3 events with 5, 10, and 20 registrants respectively	Each event shows its correct, distinct registration count with no cross-event data mixing.		